

题号

Question No.: _____

1. $X \sim \text{Uniform}[0,10] \Rightarrow \mu = \frac{1}{b-a} = \frac{1}{10}$ $Y \sim \text{Uniform}[2,8]$

$\sigma_x^2 = \text{Var}(X) = \frac{(b-a)^2}{12} = \frac{100}{12} = \frac{50}{6} = \frac{25}{3}$, $E(Y) = \frac{1}{8-2} = \frac{1}{6}$

X, Y independent

$\text{Var}(Y) = \frac{36}{12} = 3$

$E(Z) = E(X+Y) = E(X) + E(Y) = \frac{1}{6} + \frac{1}{10} = \frac{5}{30} + \frac{3}{30} = \frac{8}{30} = \frac{4}{15}$

$\sigma_z^2 = \text{Var}(Z) = \text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y) = \frac{25}{3} + 3 = \frac{34}{3}$

$\text{Cov}(X, Z) = \text{Var}(Z) - \text{Var}(X) = \frac{9}{3} = 3$

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2. A - non-singular $\Rightarrow \det(A) \neq 0$ have A^{-1}

$\Rightarrow Ax = \lambda x$

$A^{-1}(Ax) = x = A^{-1}\lambda x$
 $\Rightarrow A^{-1} = \lambda^{-1}x$ $\square$

b) $Ax = \lambda x \Leftrightarrow (A - \lambda I)^T x = 0$, Suppose eigenvector is x , corresponding

eigenvalue is λ .

$(A - \lambda I)^T x = 0 \Rightarrow (A - \lambda I)x = 0$

$(A - \lambda I)Ax = AAx = \lambda Ax$

$A = U\Sigma V^*$ $Ax = U\Sigma V^*x$
 $\uparrow$ transpose
 $\downarrow$ be longer
 $\downarrow$ transpose

$(A - \lambda I)^T = (U\Sigma V^* - \lambda I)^T x$
 $= (U(\Sigma - \lambda) V^*)^T x$
 $= (V^*(\Sigma - \lambda) U^*)^T x$
 $\downarrow$ transpose
 $\downarrow$ be longer
 $\downarrow$ transpose

$\Rightarrow$ So x is the eigenvector of A

and $(A - \lambda I)^T$

答题不得写在红线外

Do not write your answer outside the red margins.

only the location of eigenvalue is different.

题号

Question No.: 3

$$f(x, \theta) = (\theta + 1)x^\theta$$

MLE

$$\Rightarrow \prod_{i=1}^n (\theta + 1)x_i^\theta = L(\theta)$$

$$l(\theta) = \ln L(\theta) = \ln \prod_{i=1}^n (\theta + 1)x_i^\theta = \ln(\theta + 1)^n + \theta \sum_{i=1}^n \ln x_i$$

$$= \sum_{i=1}^n \theta \ln(\theta + 1)x_i$$

$$= \ln(\theta + 1) + \theta \sum_{i=1}^n \ln x_i$$

$$= \theta \sum_{i=1}^n [\ln(\theta + 1) + \ln x_i]$$

$$= \theta \sum_{i=1}^n \ln(\theta + 1)x_i$$

$$= \theta [\ln(\theta + 1) + \ln x_1] [\ln(\theta + 1) + \ln x_2]$$

$$= \theta [\ln(\theta + 1) + \ln x_1] [\ln(\theta + 1) + \ln x_2]$$

$$[\ln(\theta + 1) + \ln x_3] \dots [\ln(\theta + 1) + \ln x_n]$$

$$= \sum_{i=1}^n \theta \ln(\theta + 1)x_i = \theta \ln(\theta + 1) \sum_{i=1}^n x_i$$

$$\frac{\partial l(\theta)}{\partial \theta} = 0$$

$$\frac{\partial l(\theta)}{\partial \theta} = 0 = \ln(\theta + 1) + \theta \frac{1}{\theta + 1} = 0$$

$$\Rightarrow \theta =$$

$$\theta =$$

$$l(\theta) = \ln L(\theta) = \ln \prod_{i=1}^n (\theta + 1)x_i^\theta = \ln((\theta + 1)x_1^\theta) \ln((\theta + 1)x_2^\theta) \dots \ln((\theta + 1)x_n^\theta)$$

$$= \ln(\theta + 1) + \ln(x_1^\theta) + \dots + \ln(x_n^\theta) = \ln(\theta + 1) + \theta \sum_{i=1}^n \ln x_i$$

$$\sum_{i=1}^n \ln(\theta + 1)x_i^\theta = \sum_{i=1}^n [\ln(\theta + 1) + \ln x_i^\theta]$$

$$= \ln(\theta + 1) + \theta \sum_{i=1}^n \ln x_i = l(\theta)$$

$$\frac{\partial l(\theta)}{\partial \theta} = \frac{1}{\theta + 1} + \frac{\sum_{i=1}^n \ln x_i}{\theta + 1} = 0$$

$$\Rightarrow \theta = -\frac{1}{\theta + 1}$$

$$(\theta + 1) = -\frac{1}{\theta}$$

$$\theta = -\frac{1}{\theta + 1}$$

$$\theta = \frac{\sum_{i=1}^n \ln x_i}{\sum_{i=1}^n x_i}$$

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题号

Question No.: 4

$$P(X=1) = \frac{1}{5}$$

$$P(X=2) = \frac{2}{5}$$

$$P(X=3) = \frac{1}{5}$$

$$P(X=4) = \frac{1}{5}$$

$$P(X=5) = \frac{1}{5}$$

$$P(Y=1)$$

$$P(Y=2)$$

$$P(Y=3)$$

$$P(Y=4)$$

$$P(Y=5)$$

$$= \frac{1}{5}$$

$$\Rightarrow H(X) = -\frac{1}{5} \log_2 \frac{1}{5} \times 5$$

$$= -\log_2 \frac{1}{5}$$

$$H(Y) = -\log_2 \frac{1}{5}$$

$$H(X|Y) = \frac{1}{5} \times \left[\left(\frac{1}{5} \log_2 \frac{1}{5} + \frac{2}{5} \log_2 \frac{2}{5} + \frac{1}{5} \log_2 \frac{1}{5} \right) \right]$$

$$= \frac{1}{5} \left[-\left(\frac{1}{5} \log_2 \frac{1}{5} \right) \times 5 - \left(\frac{2}{5} \log_2 \frac{2}{5} + \frac{1}{5} \log_2 \frac{1}{5} + \frac{2}{5} \log_2 \frac{2}{5} \right) - \left(\frac{2}{5} \log_2 \frac{2}{5} + \frac{1}{5} \log_2 \frac{1}{5} \right) \times 3 - \left(\frac{2}{5} \log_2 \frac{2}{5} + \frac{2}{5} \log_2 \frac{2}{5} \right) - \left(\frac{1}{5} \log_2 \frac{1}{5} \right) \times 2 + \frac{2}{5} \log_2 \frac{2}{5} \right]$$

$$H(X|Y) =$$

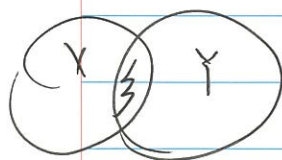
$$H(Y|X)$$

$$= \frac{1}{5} \left[-\left(\frac{1}{5} \log_2 \frac{1}{5} \right) + \left(\frac{2}{5} \log_2 \frac{2}{5} \right) \times 2 - \left(\frac{1}{5} \log_2 \frac{1}{5} \right) \times 2 + \left(\frac{2}{5} \log_2 \frac{2}{5} \right) - \left(\frac{1}{5} \log_2 \frac{1}{5} \right) \times 3 + \frac{2}{5} \log_2 \frac{2}{5} - \left(\frac{1}{5} \log_2 \frac{1}{5} \right) \times 3 + \frac{2}{5} \log_2 \frac{2}{5} - \left(\frac{1}{5} \log_2 \frac{1}{5} \right) \times 2 + \frac{2}{5} \log_2 \frac{2}{5} \right] =$$

$$I(X, Y) = H(X) - H(X|Y)$$

$$= \log_2 5 - H(X|Y)$$

+ 14



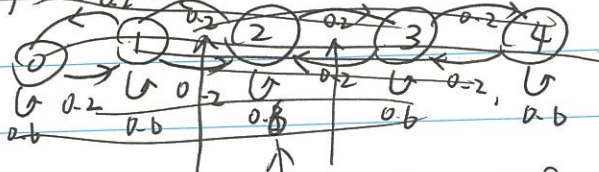
题号

Question No.: 5.

Home
2

Office
2

suppose Jim is at home



(i)

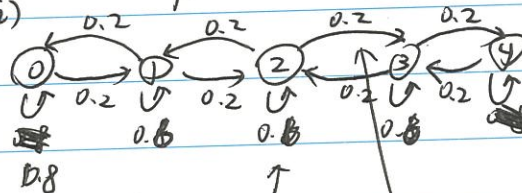
from office to bring to home.

from home bring to office

no rain

suppose at home, is the same in office.

(ii)



(iii)

can be written as

Matrices

Row Sum = 1

(iv) So we need to find P_0

$$\begin{cases} P_2 = 0.6P_2 + 0.2P_3 + 0.2P_1 \\ P_1 = 0.6P_1 + 0.2P_2 + 0.2P_0 \\ P_0 = 0.8P_0 + 0.2P_1 \\ P_3 = 0.6P_3 + 0.2P_4 + 0.2P_2 \\ P_4 = 0.8P_4 + 0.2P_3 \end{cases}$$

$$\begin{matrix} & 0 & 1 & 2 & 3 & 4 \\ \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0.8 & 0.2 & 0 & 0 & 0 \\ 0.2 & 0.6 & 0.2 & 0 & 0 \\ 0 & 0.2 & 0.6 & 0.2 & 0 \\ 0 & 0 & 0.2 & 0.6 & 0.2 \\ 0 & 0 & 0 & 0.2 & 0.8 \end{bmatrix} \end{matrix}$$

rain today

$$P_0 = 0.2 \times 0.2 \times 0.2 = (0.2)^3$$

from 1 to 0

from 2 to 1

-2

题号

Question No.:

6

$$D_n^{(\alpha, \beta)} = \sum$$

$$D(p||q) = \sum p \log \frac{p}{q}$$

$$p(x) = \{1, 0, 0, \dots, 0\}$$

$$q(x) = \left\{1 - \left(\frac{\log \alpha}{\log n}\right)^\beta, \frac{1}{n} \left(\frac{\log \alpha}{\log n}\right)^\beta, \dots, \frac{1}{n} \left(\frac{\log \alpha}{\log n}\right)^\beta, 0, \dots, 0\right\}$$

$$n \rightarrow \infty \quad \beta \rightarrow 0$$

$$D_n^{(\alpha, \beta)} \rightarrow \frac{1}{n} \left(\frac{\log \alpha}{\log n}\right)^\beta$$

$$\Rightarrow \lim_{n \rightarrow \infty} 0 \cdot \log \frac{0}{\frac{1}{n} \left(\frac{\log \alpha}{\log n}\right)^\beta} = 0 \cdot \log \frac{0}{0} = 0$$

$$\lim_{n \rightarrow \infty} \frac{\frac{1}{n} \left(\frac{\log \alpha}{\log n}\right)^\beta}{n} = \frac{0}{\infty} = 0$$

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